

Weijun Li 利伟君

Postdoctoral Researcher · Institute of High Energy Physics

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Experimental neutrino physics · Detector reconstruction · Statistical inference

Appointments

February 2026–present

Postdoctoral Researcher · Institute of High Energy Physics, China

Machine learning for atmospheric neutrino reconstruction at JUNO.

2020–2021

Quantum Computing Associate · Horizon Quantum Computing, Singapore

Implementation of quantum signal processing in MATLAB.

2017–2019

Research Assistant · Centre for Quantum Technologies, Singapore

Monte Carlo sampling, Bayesian inference and quantum process tomography.

Education

2021–2025

DPhil in Particle Physics · University of Oxford

Thesis: Next-Generation Neutrino Interaction Measurements and Beyond-Standard-Model Physics with the Upgraded T2K Near Detector.

2019–2020

MASt in Physics · University of Cambridge

First class.

2013–2017

BSc (Hons) in Physics · National University of Singapore

Highest Distinction.

Grants and awards

2022–2023 · Great Britain Sasakawa Foundation Grant · Great Britain Sasakawa Foundation

2022 · Denys Wilkinson Prize · University of Oxford

2020 · Bateman Scholar · Trinity Hall, University of Cambridge

2013–2017 · Science & Technology Undergraduate Scholarship · National University of Singapore

2013–2017 · Dean's List (five semesters) · National University of Singapore

2015 · CRISP Award · National University of Singapore

2015 · Silver Medal · University Physics Competition

Publications and preprints

Sampling Quantum States with Inequality Constraints

Weijun Li, Rui Han, Jiangwei Shang, Hui Khoon Ng, Berthold-Georg Englert

Entropy 28, 614 (2026) · Published journal article

doi:10.3390/e28060614 · arXiv:2109.14215

New Physics Opportunities at Neutrino Facilities: BSM Physics at Accelerator, Atmospheric, and Reactor Neutrino Experiments

Koun Choi, Doojin Kim, Jong-Chul Park, et al. (including Weijun Li)

arXiv (2025) · Preprint

arXiv:2506.15306

Center-of-momentum variables in charged-current single-pion–single-proton events

Weijun Li

Physical Review D 111, 072010 (2025) · Published journal article

doi:10.1103/PhysRevD.111.072010

First combined tuning on transverse kinematic imbalance data with and without pion production constraints

Weijun Li, Marco Roda, Júlia Tena-Vidal, et al.

Physical Review D 110, 072016 (2024) · Published journal article

doi:10.1103/PhysRevD.110.072016 · arXiv:2404.08510

Uncorrelated problem-specific samples of quantum states from zero-mean Wishart distributions

Rui Han, Weijun Li, Shrobona Bagchi, Hui Khoon Ng, Berthold-Georg Englert

arXiv (2021) · Preprint

arXiv:2106.08533

Very strong evidence in favor of quantum mechanics and against local hidden variables from a Bayesian analysis

Yanwu Gu, Weijun Li, Michael Evans, Berthold-Georg Englert

Physical Review A 99, 022112 (2019) · Published journal article

doi:10.1103/PhysRevA.99.022112

A General Free Plane Wave Ansatz to The Classcal SU(2) Yang-Mills Theory: with Application to Gravity

W. Li, A. H. Chan, C. H. Oh

EPJ Web of Conferences 206, 09011 (2019) · Conference proceedings

doi:10.1051/epjconf/201920609011

Public software

QSCMC · Sequentially constrained Monte Carlo sampling of quantum states.

QSam · Problem-specific, uncorrelated quantum-state sampling.

Talks, posters and experience

21 June 2024 · Poster · presenter

Transverse Kinematic Imbalance Analysis and Pion Trackless Reconstruction at the upgraded T2K near detector

Neutrino 2024 · XXXI International Conference on Neutrino Physics and Astrophysics

Conference record and poster

18 June 2024 · Poster · coauthor

First combined analysis of Transverse Kinematic Imbalance data with and without pion production constraints

Neutrino 2024 · XXXI International Conference on Neutrino Physics and Astrophysics

Conference record

4 June 2024 · Plenary talk

BSM at T2SK/HK near detectors

4th Workshop on New Physics Opportunities at Neutrino Facilities

Conference record

4 June 2024 · Contributed talk

BeamHNL, a GENIE-based general heavy neutral lepton generator

4th Workshop on New Physics Opportunities at Neutrino Facilities

26 October 2023 · Parallel talk

Assembly, test and analysis development of the T2K upgrade

XX International Workshop on Neutrino Telescopes

Conference record and slides · Zenodo record

5 July 2023 · Parallel talk

T2K latest neutrino oscillation results

WIN 2023 · 29th International Workshop on Weak Interactions and Neutrinos

Conference record and slides

8 March 2023 · Seminar

Recent Oscillation Results and Upgrade of the T2K Experiment

SPeCial4Young · SYSU–PKU Collider Physics Forum for Young Scientists

Seminar record

3–7 September 2018 · Poster

A general free plane wave ansatz to the classical SU(2) Yang–Mills theory: with application to gravity

XLVIII International Symposium on Multiparticle Dynamics

Related proceedings

Collaborations and detector work

GENIE (since July 2023); T2K (since October 2021). SuperFGD assembly and hardware work; training and work as a data-acquisition expert.

Scientific computing and languages

C++, Python, MATLAB, LaTeX and Linux.

Cantonese and Mandarin Chinese (native); English (proficient); German (intermediate); Japanese (beginner).